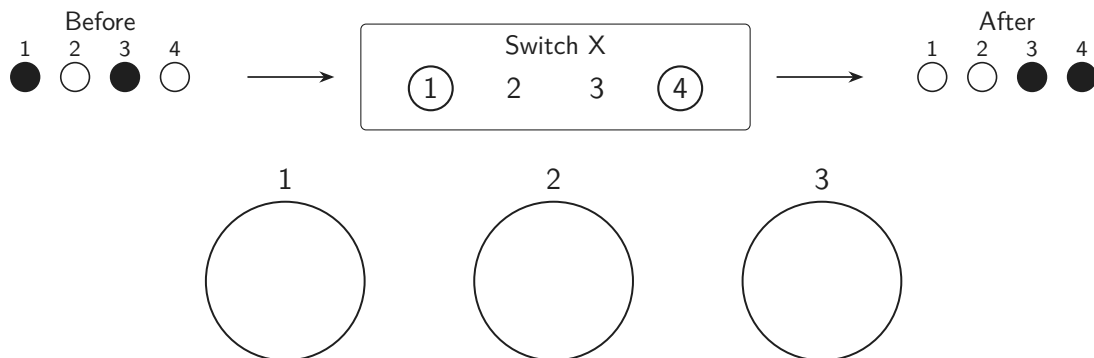


Choose ON and OFF faces for the counters. In pictures, ● = ON and ○ = OFF.

Problem 1: Your partner circles none, some, or all lamp numbers on each switch card and hides the cards. The cards stay fixed during a round. Start with all lamps OFF. Call one or more switches. For each switch you call, your partner flips every lamp named on its card. Find what A, B, and C change. Shade the ON lamps in your trial records, then trade jobs with new cards.



Switches called	Before	After
1 ☐ A ☐ B ☐ C	1 2 3 ○ ○ ○	1 2 3 ○ ○ ○
2 ☐ A ☐ B ☐ C	1 2 3 ○ ○ ○	1 2 3 ○ ○ ○
3 ☐ A ☐ B ☐ C	1 2 3 ○ ○ ○	1 2 3 ○ ○ ○
4 ☐ A ☐ B ☐ C	1 2 3 ○ ○ ○	1 2 3 ○ ○ ○
5 ☐ A ☐ B ☐ C	1 2 3 ○ ○ ○	1 2 3 ○ ○ ○

A changes: _____ B changes: _____ C changes: _____

Switch A

1 2 3

Switch B

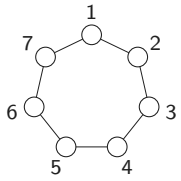
1 2 3

Switch C

1 2 3

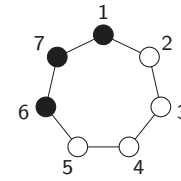
Problem 2: Start each target with all lamps OFF and all switch numbers uncircled. Call a switch number; your partner flips that lamp and the next two lamps clockwise. Make each target picture, or cross it out if you think it cannot be made. Circle each switch number when you press it; erase the circle if you press it again.

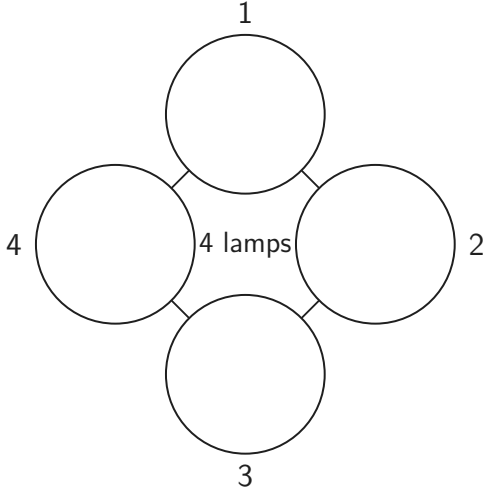
Before: 7 lamps



Press 6
flip 6, 7, 1

After





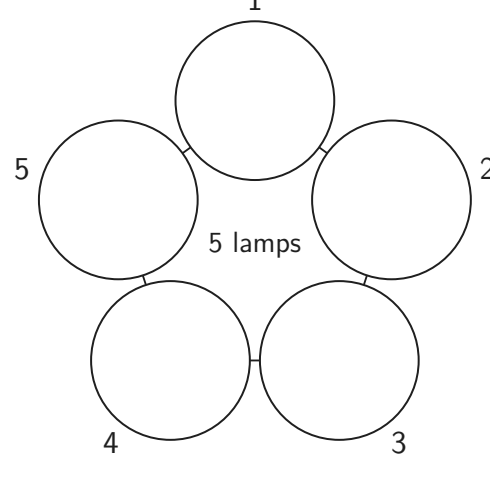
4 lamps

Target: 1 (black), 2 (white), 3 (white), 4 (white)

Switches: _____

Target: 1 (black), 2 (black), 3 (black), 4 (black)

Switches: _____



5 lamps

Target: 1 (black), 2 (white), 3 (white), 4 (white), 5 (white)

Switches: _____

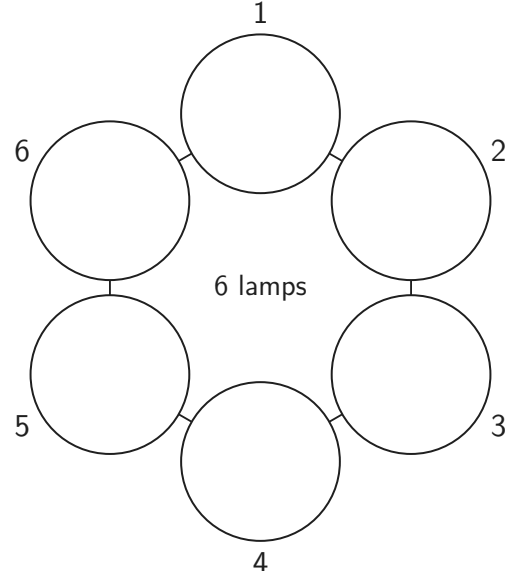
Target: 1 (black), 2 (black), 3 (black), 4 (black), 5 (black)

Switches: _____

Target

1 (black), 2 (white), 3 (white), 4 (white), 5 (white), 6 (white)

Switches: _____



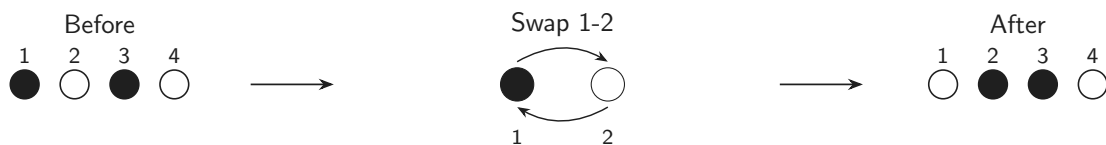
6 lamps

Target

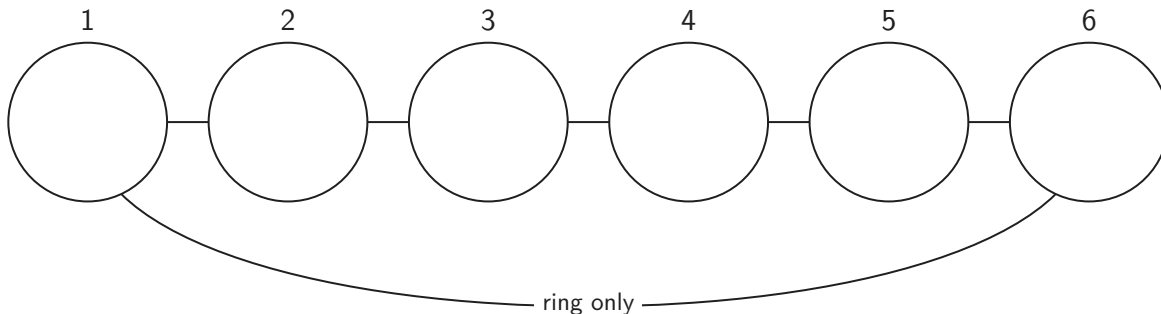
1 (black), 2 (black), 3 (black), 4 (black), 5 (black), 6 (black)

Switches: _____

Problem 3: A switch now swaps the two counters at the ends of a join. Your partner makes the swap. Change each BEFORE into its TARGET with as few swaps as you can. Start from BEFORE for both the line and ring. Record the joins you swap. Cross out a target if it cannot be made.

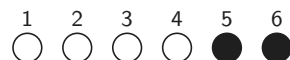
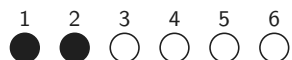


Line: straight joins only. Ring: also use the curved join.



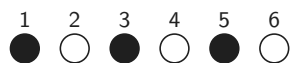
Line joins: _____

Ring joins: _____



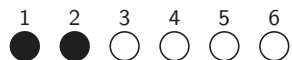
Line joins: _____

Ring joins: _____



Line joins: _____

Ring joins: _____



Line joins: _____

Ring joins: _____